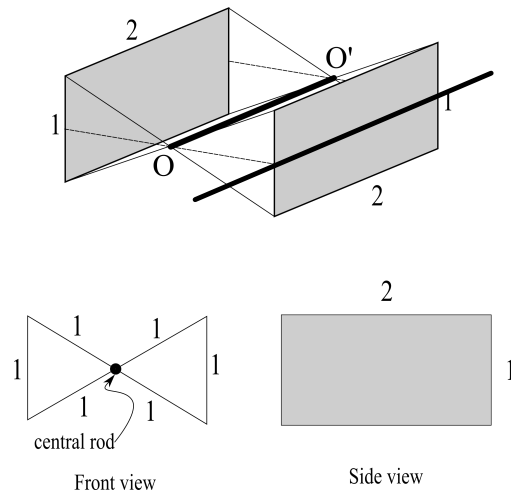


AERO40007 – Mechanics

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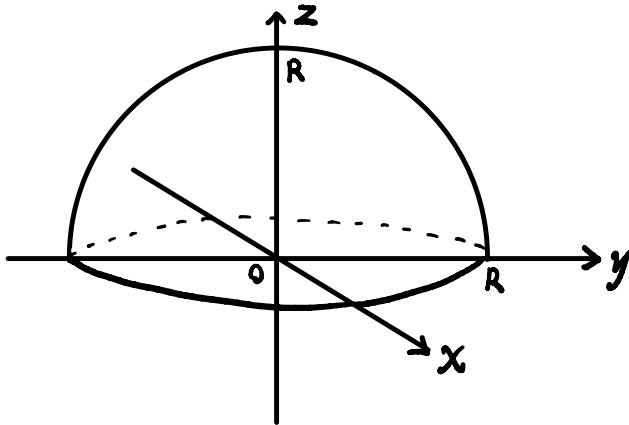
Centres of mass and moments of inertia

(1.) The displayed structure is made of two panels of thickness 1mm and density 7680 kg/m^3 . They are attached to the central rod by thin bars of negligible mass. If the central rod has a moment of inertia with respect to OO' of I_1 . Find the total moment of inertia of the structure with respect of the axis OO' . All dimensions in the figure are in m.



$$\begin{aligned}
 I_{OO} &= I_1 + (I_G + d^2 M) \times 2 \\
 &= I_1 + \left[\frac{1}{12} b h^3 + d^2 M \right] \times 2 \\
 &= I_1 + \left[\frac{1}{12} \times 2 \times 1^3 \times (0.001) \times 7680 + \left(\frac{\sqrt{2}}{2} \right)^2 \times 1 \times 2 \times (0.001) \times 7680 \right] \times 2 \\
 &= I_1 + (1.28 + 11.52) \times 2 \\
 &= I_1 + 25.6
 \end{aligned}$$

(2.) Locate the centroid of the volume of a hemisphere (half sphere) of radius R with respect to its base.



$$x^2 + y^2 + z^2 = R^2$$

$$x^2 + y^2 = R^2 - z^2$$

$$\bar{V} = \frac{1}{V} \int z \, dV$$

$$= \frac{1}{V} \int_0^R \pi (x^2 + y^2) z \, dz$$

$$= \frac{1}{V} \int_0^R \pi (R^2 - z^2) z \, dz$$

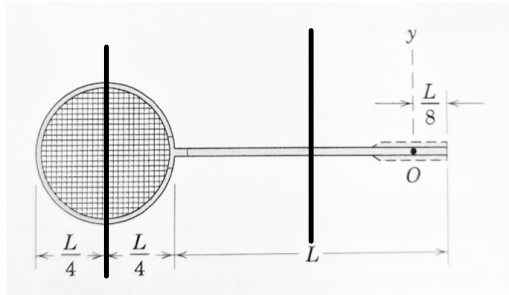
$$= \frac{\pi}{\frac{1}{2} \times \frac{4}{3} \pi R^3} \int_0^R R^2 z - z^3 \, dz$$

$$= \frac{1}{\frac{2}{3} R^3} \left[\frac{1}{2} R^2 z^2 - \frac{1}{4} z^4 \right]_0^R$$

$$= \frac{3}{2R^3} \left(\frac{1}{2} R^4 - \frac{1}{4} R^4 \right)$$

$$= \frac{3}{8} R$$

(3.) A new design for badminton rackets has a circular head and a straight shaft and handle. All made with the same slender rod. Find the moment of inertia with respect to y axis. The rod material has a linear density of ρ . Neglect the strings and wooden handle.



$$I_y = I_{cm} + d^2 M + I_o + d^2 M$$

$$= \frac{1}{3} m \left(\frac{L}{2} \right)^2 + \left(\frac{3}{8} L \right)^2 m + \frac{1}{2} m \left(\frac{L}{4} \right)^2 + \left(\frac{9}{8} L \right)^2 m$$

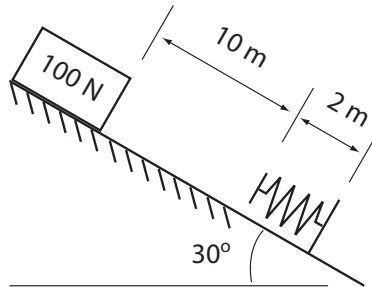
$$= \frac{1}{3} (\rho L) \left(\frac{L^2}{4} \right) + \frac{9}{64} L^2 (\rho L) + \frac{1}{2} (\rho \pi \frac{L}{2}) \left(\frac{L^2}{16} \right) + (\rho \pi \frac{L}{2} \times \frac{81}{64} L^2)$$

$$= \frac{1}{12} \rho L^3 + \frac{9}{64} \rho L^3 + \frac{1}{64} \rho \pi L^3 + \frac{81}{128} \rho \pi L^3$$

$$= \frac{43}{142} \rho L^3 + \frac{83}{128} \rho \pi L^3$$

Energy and Momentum

(4.) A 100 N block is placed on a 30° slope with coefficient of friction $\mu = 0.3$ in the position shown. The block is then released. What is the maximum distance the spring, stiffness $k = 10\text{ N/mm}$ will be compressed



$$E = mgh - fd = \frac{1}{2} k \Delta L^2$$

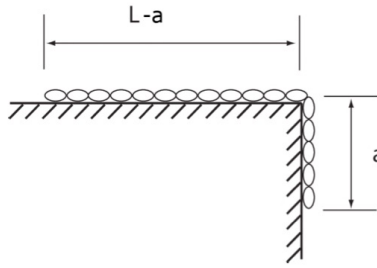
$$\therefore 100 \sin 30^\circ (10 + \Delta L) - 100 \cos 30^\circ \cdot 0.3 \cdot (10 + \Delta L) = \frac{1}{2} \cdot 10000 \cdot \Delta L^2$$

$$\therefore 500 + 50\Delta L - 150\sqrt{3} - 15\sqrt{3}\Delta L = 5000\Delta L^2$$

$$\therefore 5000\Delta L^2 - (50 - 15\sqrt{3})\Delta L + (150\sqrt{3} - 500) = 0$$

$$\therefore \Delta L = 0.22159\text{ m} = 0.222\text{ m}$$

(5.) A chain of length L is released from rest on a frictionless surface as shown. Determine the velocity when the last link moves off the horizontal surface.



$$\frac{1}{2}mv^2 = \Delta mgh$$

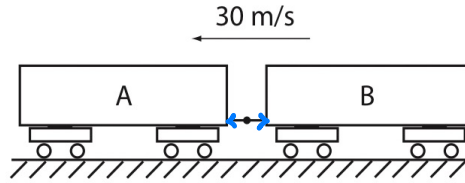
$$\therefore \frac{1}{2}\rho L v^2 = \rho L g \left(\frac{L}{2}\right) - \rho a g \left(\frac{a}{2}\right)$$

$$\therefore \frac{1}{2}L v^2 = \frac{1}{2}g L^2 - \frac{1}{2}g a^2$$

$$\therefore v^2 = gL - g\frac{a^2}{L} = g\left(\frac{L^2 - a^2}{L}\right)$$

$$\therefore v = \left(\frac{g(L^2 - a^2)}{L}\right)^{\frac{1}{2}}$$

(6.) A train has two carriages travelling at 30 m/s. Carriage A weighs 15 tonnes, carriage B weighs 20 tonnes. Each carriage can exert a constant braking force of 20 kN. Calculate the time required to stop the train, and the force in the coupling between the carriages, when the brakes in both carriages are applied.



$$a_{\text{total}} = \frac{15T + 20T}{2 \times 20kN} = \frac{8}{7} \text{ m/s}^2$$

$$\therefore t = \frac{v}{a} = 26.25 \text{ s}$$

$$F_t = ma = (20T - 15N) \times \frac{8}{7} \text{ m/s}^2 = \frac{40}{7} \text{ kN}$$

$$\therefore F = \frac{1}{2} F_t = \frac{20}{7} \text{ kN} = 2.86 \text{ kN}$$

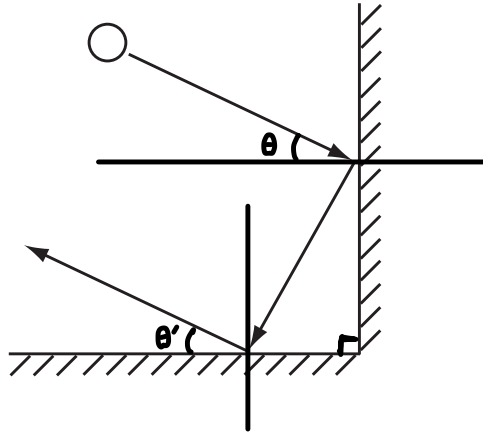
OR

$$\frac{-F - 20 \text{ kN}}{15T} = \frac{F - 20 \text{ kN}}{20T}$$

$$\therefore F = -\frac{20}{7} \text{ kN}$$

$$\therefore F = \frac{20}{7} \text{ kN}$$

(7.) A ball is thrown into a 90° corner as shown with initial velocity v . The coefficient of restitution between the ball and the wall is ϵ . Show that (i) the final velocity of the ball is ϵv and (ii) the final path of the ball is parallel to the initial path.



$$(1) \quad \textcircled{1} \quad x\text{-direction} : \frac{-v'_x}{v \cos \theta} = \epsilon \quad \therefore v'_x = -\epsilon v \cos \theta$$

y -direction do not change

$\textcircled{2} \quad x$ -direction do not change

$$y\text{-direction} : \frac{-v'_y}{v \sin \theta} = \epsilon \quad \therefore v'_y = -\epsilon v \sin \theta$$

$$\therefore v = \sqrt{(v'_x)^2 + (v'_y)^2} = \epsilon v$$

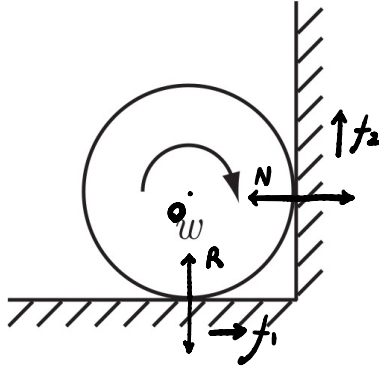
$$(ii) \quad \therefore v'_x = -\epsilon v \cos \theta$$

$$v'_y = -\epsilon v \sin \theta$$

$$\therefore \theta' = \arctan\left(\frac{v'_y}{v'_x}\right) = \theta$$

$\therefore$ parallel

(8.) A cylinder of mass m and radius r is placed in a 90° corner with clockwise angular velocity ω . The coefficient of friction between the cylinder and the walls is μ . Derive an expression for the time required for the cylinder to come to rest.



$$F = ma = m \frac{\Delta v}{\Delta t} = m \frac{\Delta \omega r}{\Delta t}$$

$$\uparrow R + f_2 = mg = R + \mu N$$

$$\rightarrow f_1 = N = \mu R$$

$$\therefore R + \mu^2 R = mg$$

$$\therefore R = \frac{mg}{1 + \mu^2} \quad \therefore N = \frac{\mu mg}{1 + \mu^2}$$

$$\therefore f_1 = \frac{\mu mg}{1 + \mu^2}$$

$$f_2 = \frac{\mu^2 mg}{1 + \mu^2}$$

$$\therefore M(0), \int M dt = \Delta H$$

$$\therefore \int f_1 + f_2 dt = \Delta L \omega$$

$$\therefore (f_1 + f_2) t = L \omega = \frac{1}{2} m r^2 \omega \quad \therefore t = \frac{r \omega (1 + \mu^2)}{2 g \mu (1 + \mu)}$$

Answers

1.

$$25.6 + I_1 \text{ kg m}^2$$

2.

$$\frac{3}{8}R$$

3.

$$\rho L^3 \left(\frac{43}{192} + \frac{83\pi}{128} \right)$$

4.

$$\text{compression} = 0.222 \text{ m}$$

5.

$$v = \sqrt{\frac{g}{L}(L^2 - a^2)}$$

6.

stopping time = 26.25 s, force = 2.86 kN (compression)

8.

$$t = \frac{r\omega(1 + \mu^2)}{2g\mu(1 + \mu)}$$